

Department of Electrical and Computer Engineering
COMSATS University Islamabad, Attock Campus

Lab Rubrics Form	
Lab Name	String Manipulation Using User-Defined Functions in C
Lab Number	11
Submitted by:	
Student Names	Registration #
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Rubrics		Marks		
		1	2	3
PLO-5 Modern Tools Usage	Ability to execute the experiment utilizing hardware, software tools, or a combination of both.			
PLO-10 Communication Skills	Present lab activities effectively through detailed lab reports, oral examinations, and engaging presentations.			

Lab 11: String Manipulation Using User-Defined Functions in C

Objective

To understand string manipulation in C programming by implementing user-defined functions for string concatenation, comparison, case conversion, and reversal without relying on standard library string functions where specified.

Software Requirements

- Code::Blocks IDE / Dev-C++
- GCC Compiler
- Windows Operating System

Experiments Performed

In-Lab Tasks

1. Concatenate two strings using a user-defined function without using standard library string functions.
2. Compare two strings using a user-defined function and determine their length if they are identical.

Post-Lab Tasks

1. Convert a string to lowercase using the standard library function and a user-defined approach.
2. Implement a user-defined version of the `strrev()` function to reverse a string and perform its dry run.

Learning Outcomes

After completing this lab, the student will be able to:

- Understand the representation of strings in C.
- Manipulate strings using arrays of characters.
- Develop user-defined functions for common string operations.
- Compare, concatenate, reverse, and convert strings.
- Strengthen programming skills by avoiding built-in string functions where required.

In-Lab Tasks:

Task1:

Write a function to concatenate two strings without using standard library functions.

Code:

```
#include <stdlib.h>

void concatStrings(char str1[], char str2[] ) {
    int i, j;
    for (i=0; str1[i] != '\0'; i++);
    for (j=0; str2[j] != '\0'; i++, j++) {
        str1[i] = str2[j];
    }
    str1[i] = '\0';
}

int main() {
    char str1[100] = " The Hello, ";
    char str2[] = "world Life!";
    concatStrings(str1, str2);
    printf("Concatenated string: %s\n", str1);
    return 0;
}
```

Output:



Concatenated string: The Hello, world Life!

Process returned 0 (0x0) execution time : 0.125 s
Press any key to continue.

Task2:

Write a function to compare two strings without using standard library functions. If the strings are same, find the length of those strings. Make the flowchart for this task.

Code:

```
#include <stdlib.h>
int compareStrings(char str1[ ], char str2[]) {
    int i;
    for (i = 0; str1[i] != '\0' && str2[i] != '\0'; i++)
    {
        if (str1[i] != str2[i]) {
            return 0;    /*Strings are different*/
        }

        if (str1[i] == '\0' && str2[i] == '\0') {
            return i;    /*Strings are the same, return length*/
        }
        return 0;    /*Strings are different*/
    }
}

int main() {
    char str1[] = "Hello";
    char str2[] = "Hello";
    int length;
    length = compareStrings(str1, str2);

    if (length > 0) {
        printf("Strings are the same.\n");
        printf("Length: %d\n", length);
    } else {
        printf("Strings are different.\n");
    }
    return 0;
}
```

Output:A screenshot of a Windows command prompt window. The title bar shows the file path: "C:\Users\Hamza Computer\Documents\PF\lab1\ti8uf8\bin\Debug\ti8uf8.exe". The command prompt has a black background with white text. The output of the program is displayed as follows: "Strings are the same." followed by "Length: 5" on the next line. Below that, it says "Process returned 0 (0x0) execution time : 0.125 s" and "Press any key to continue." at the bottom.

Post-Lab Tasks:

Task1:

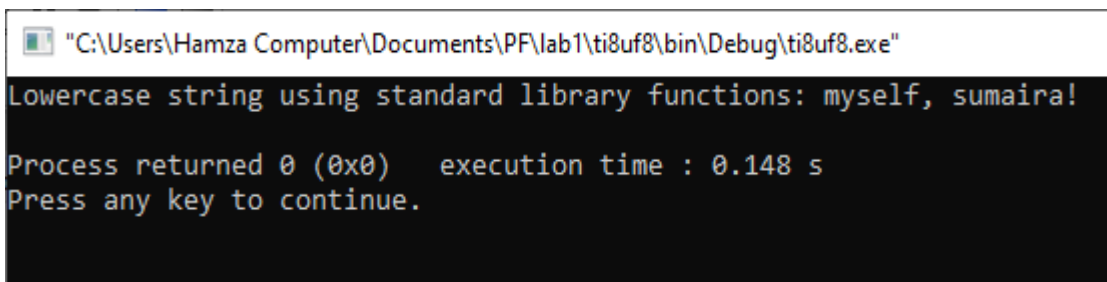
Convert string to a lower case through both standard library and user defined functions.

Code:

```
#include <stdlib.h>
void toLowerCaseStandard(char str[] )
{
    int i;
    for (i = 0; str[i] != '\0'; i++)
    {
        str[i] = tolower(str[i]);
    }
}
int main()
{
    char str[] = "Myself, Sumaira!";

    /*Convert string to lowercase using standard library functions*/
    toLowerCaseStandard(str);
    printf("Lowercase string using standard library functions: %s\n", str);
    return 0;
}
```

Output:



```
"C:\Users\Hamza Computer\Documents\PF\lab1\ti8uf8\bin\Debug\ti8uf8.exe"
Lowercase string using standard library functions: myself, sumaira!
Process returned 0 (0x0) execution time : 0.148 s
Press any key to continue.
```

Task2:

Write a user defined function of `strrev ( )` i.e. library function to reverse a string. Take a string of length five to reverse and do the dry run for it.

Code:

```
void strrev(char str[ ] )
{
    int length = 0, i, j;
    while (str[length] != '\0') {
        length++;
    }
    /*Swap characters to reverse the string*/
    for (i=0, j=length-1; i<j; i++, j--)
    {
        char temp = str[i];
        str[i] = str[j];
        str[j] = temp;
    }
}

int main() {
    char str[] = "Sumaira";
    printf("Original string: %s\n", str);
    /*Reverse the string using user-defined strrev function*/
    strrev(str);
    printf("Reversed string: %s\n", str);
    return 0;
}
```

Output: